MILITARY SPECIFICATION

REGULATOR ASSEMBLY: F11747171

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Regulator Assembly F11747171 (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Material, General Specification Governing the Manufacture and Inspection of
MIL-F-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental test methods

DRAWINGS

U.S. Army, Frankford Arsenal

F11747171	Regulator Assembly

PACKAGING DATA SHEET

PS 11747171	Regulator Assembly

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The Regulator shall be manufactured in accordance with Drawing F11747171 and all drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Performance-Environmental.- Specified ambient temperature extremes shall be within ±5°F. The Regulator shall show no evidence of damage after exposure to the following environments.

3.3.1 Storage temperatures.- The Regulator shall meet the requirements of 3.4 at ambient temperature + 75° ± 15°F after exposure and thermal stabilization at -80°F and +160°P.

3.3.2 Operating temperatures.- Unless otherwise specified, the Regulator shall meet the. requirements of 3.4 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which it shall meet the above requirements at ambient temperatures +75° ± 15°F.

3.3.3 Shock.- The Regulator shall meet the requirements of 3.4 after being subjected to cumulative shocks consisting of 3 each at 120 G’s and 50 G’s applied in each direction along each of three mutually perpendicular axes. The shock pulse shall have a wave form of approximately a half sine wave. For 120 G's ± 10%, the shock pulse shall have a time duration of 3 ± .2 nilliseconds; for 50 G’s ± 10% the shock pulse shall have a time duration of 18 ± 3 milliseconds.

3.3.4 Vibration “A”.- The Regulator shall meet the requirements of 3.4 after being vibrated sinusoidally along each of three mutually perpendicular axes as follows:

(a) At a frequency of 5 cycles per second (cps) increase acceleration linearly from 0 to .65 G 's (peak).
(b) At a frequency of 5 to 9 cycles per second (cps) increase acceleration linearly from .65 to 1.5 G’s (peak),
(c) At a frequency of 9 to 500 cycles per second (cps) at a constant acceleration of 1.5 G’s (peak).

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(d) At a frequency of 500 cycles per second (cps) decrease acceleration linearly from 1.5 to O G's (peak). The frequency of applied vibration shall be swept logarithmically in 18 min ±1 minute sweep cycles for a total of two hours per axis.

3.3.5 Vibration "B".- The Regulator shall meet the requirements of 3.4 after being vibrated in a vertical plane at a constant frequency of 30 cycles per second (CPS) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds.

3.4 Performance-Operational.- The Regulator assembly shall meet the following requirements with loads applied as shown in figure 1 and energized with direct current (de) electrical potential from 18.0 to 30.0 volts. NOTE: all pin numbers refer to connector MS24006-3.

3.4.1 Polarization of input terminal.- The regulator input terminal shall be polarized positive pin Pi-D with respect to the Regulator pin P1-F, which shall be negative and considered as electrical circuit ground.

3.4.2 Polarization of output terminal.- The regulator output terminal pin PI-B shall deliver an electrical potential of positive polarity with respect to the electrical circuit ground.

3.4.3 Reverse polarity.- The regulator shall not be damaged by the application of electrical potentials of 3.4 in reverse polarity.

3.4.4 Intermittent operation- The regulator shall meet the following requirements after being subjected to one hour of intermittent operation consisting of a continuously repetitive 30 second ON and 30 second OFF cycle at an ambient temperature of +75° ± 15°F.

3.4.4.1 Input voltage surge.- The regulator shall show no evidence of damage and the output voltage shall remain within 2.0 to 3.0 volts during the application of 100 millisecond positive square wave pulses having an amplitude of 80.0 ± 1.0 DC volts superimposed on input voltage. The output voltage shall return to the level specified in 3.4.4.3 within 125 milliseconds after removal of pulses.

3.4.4.2 Output current.- The Regulator shall deliver a minimum of 500.0 dc milliamperes when energized by the potentials of 3.4.

3.4.4.3 Output voltage.- The Regulator shall deliver a dc voltage of 2.35 ± .35 volts with a rise time of 0.3 seconds or less upon opening test shutter switch.

3.4.4.4 Shutter switch operation.- The Regulator shall provide output current and voltage as specified in 3.4.4.2 and 3.4.4.3 only after the test shutter switch Pi-E to ground is positioned from the closed to the open position with input power applied. The Regulator shall not provide an output when the test shutter switch is in the closed position, or when in the open position while input power is being applied.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in M1L-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 10 Regulators manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and test.- Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristics) basis in accordance with MIL-STD-105 and plans specified in Tables I, II, III, IV and V herein. Examinations and tests for packaging, packing and marking shall be in accordance with .

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MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III, IV and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I

100% Inspection

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Polarization of input terminal	3.4.1	4.5.7.1
102.	Polarization of output termina1	3.4.2	4.5.7.2
103.	Reverse polarity	3.4,3	4.5.7.3
104.	Intermittent operation	3.4*4	4.5.7.4
105.	Input voltage surge	3.4.4.1	4.5.7,4.1
106.	Output current	3.4.4.2	4.5.7.4.2
107.	Output voltage	3.4.4.3	4.5.7.4.3
108.	Shutter switch operation	3.4.4.4	4.5.7.4.4

TABLE II

CLASS
CRITICAL: NONE
MAJOR: AQL 1.0 % Defective

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
109.	Vibration "B"	3.3.5	4.5.5

MINOR: NONE

NOTE: Samples shall meet the requirement of Table I subsequent to Vibration "B" test.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defecrive units and resubmittance of rejected lots shall be in accordance with " "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Special sampling.

4.3.4.1 Two assemblies shall be selected at random as a special sampling from each 100 produced, or from each months production whichever occurs first. The sample shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings-visual
302.	General specifications	3.2	MIL-F-1392& visual

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4.3.4.2 Three assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall then meet the requirements and tests in Table I.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature	3.3.1	4.5.1
304.	Operating temperature	3.3.2	4.5.2
305.	Shock	3.3.3	4.5.3

4.3.4.3 Three assemblies shall be selected at random as samples from production lots at the 25 percent, 50 percents and 75 percent completion interval of each production order or contract. Each sample shall meet the requirements and tests in Table V and shall then meet the requirements and tests in Table I.

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
306.	Vibration "A"	3.3.4	4.5.4

4.3.4.4 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made-by the contractor and the resubmitted items have been approved (see 4.3.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-1-45607.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with 4.4.2.3, the "Test Methods and Procedures" specified in 4.5 of this specification and with the component inspection-procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be consi-

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dered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming, Construction shall be such as to facilitate routine calibration of test equipment.

4.4.2.3 Inspection equipment requirements.

4.4.2.3.1 Temperature test.- The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F. The temperature gradient across the cross-sectional area occupied by the test items shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radiant heat, if used, will not fall directly on the test item. Therocouples or equilent temperatue sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber, or in the return air stream and shall be baffled or otherwise protected against direct impingement of air supply and against radiation effects.

4.4.2.3.2 Shock test.- The shock testing apparatus shall conform to the requirements of MIL-STD-810 for half sine wave shock impulse testing.

4.4.2.3.3 Vibration Test.- The vibration equipment shall provide the specified amplitudes within plus or minus 5 percent, the frequencies within plus or minus 2 percent and the accelerations within plus or minus 10 percent. The vibration equipment for vibration "A" test shall be capable of vibrating regulator along three mutually perpendicular axes while the Regulator is maintained in its' normal mounting or operating position (as depicted in the front view of F11747171). Orientation of regulator for Vibration "B" test shall also be in its' normal or operating position. Adapters shall be provided to properly secure regulator to vibration testers,

4.4.2.3.4 Operational tests.- Test equipment listed in the following Table VI shall be utilized in performing applicable tests of 4.5:

TABLE VI - TEST EQUIPMENT LIST

ITEM NO.	DESCRIPTION
1. Armeter	D.C., 0 - 1A, 1% Accuracy
2. D.C. Power Supply	Plus 18 + 1.0 volts Plus 30 - 1.0 volts Regulated to 1% Ripple 1% RMS Max.
3. D.C. Power Supply	Plus 80 ± 1.0 volts Regulated to 1% Ripple 1% RMS Max.
4. Oscilloscope	Bandwidth 10MHZ Min deflection 10mv/div Max sweep 50 ns

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TABLE VI - TEST EQUIPMENT LIST CON'T

ITEM NO.		DESCRIPTION
5.	Pulse Generator	Positive square wave, Pulse width 100 msec- triggering capability of 4.5.7.4.1. Use in conjunction with item 3
6.	Voltmeter	D.C., O to 10v, accuracy of 1%.

4.5 Test methods and procedures,

4.5.1 Storage temperatures.- Testing equipment utilized in this test shall conform to the requirements of 4.4.2.3.1. The Regulator shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature to -80°F. The Regulator shall be maintained at this extreme temperature for a minimum of 4 hours. Upon competion of the 4 hour exposure the temperature of the chamber shall be raised, over a period of 8 hours plus or minus 15 minutes, to + 160°F. The regulator shall remain at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure the temperature of the chamber shall be reduced over a period of 4 hours plus or minus 15 minutes to room ambient temperature. The Regulator shall then be removed from the test chamber and subjected to the visual and tactile examination of 4.5.6. The Regulator shall conform to the requirements of 3.3.1. Caution shall be exercise during test to avoid sub-jecting regulator to thermal shock.

4.5.2 Operating temperatures.- Testing equipment utilized in this test shall conform to the requirements of 4.4.2.3.1. The Regulator shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room temperature to -40°F. The Regulator shall be maintained at theis extreme temperature for a minimum of 4 hours. At completion of the 4 hour exposure, and while at -40°F, the Regulator shall be subjected to the tests in 4.5.7. Upon completion of the low temperature operability tests the temperature of the chamber shall be raised, over a period of 8 hours plus or minus 15 minutes, to +125°F. The Regulator shall be maintained at this extreme temperature for a minimum of 4 hours. At the completion of the 4 hour exposure, and while at +125°F the Regulator shall be subjected to the tests in 4.5.7. Upon completion of the high temperature operability tests the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The Regulator shall then be removed from the test chamber. The Regulator shall conform to the requirements of 3.3.2 at each temperature extreme. Caution shall be exercised during test to avoid subjecting regulator to thermal shock.

4.5.3 Shock.- The test equipment utilized in this test shall conform to the requirements of 4.4.2.3.2. The Regulator shall be shock tested in accordance with Test Method 516.2, Procedure I with a half sine wave shock pulse of MIL-STD-810, except conditions such as "G" values and time

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duration shall be as specified in 3.3.3. Upon completion of the 36 shock impulses the Regulator shall be removed from the shock test machine and adapter and subjected to the examinations in 4.5.6. Regulators that have passed the examinations specified in 4.5.6 shall then be subjected to the tests specified in 4.5.7 through 4.5.7.4.4 inclusive, to determine overall conformance to 3.3.3.

4.5.4 Vibration "A".- Testing equipment utilized in this test shall conform to the requirements of 4,4.2.3.3. The Regulator shall be vibrated in accordance with Test Method 514.2, Prodceure VIII (tracked Vehicles) of MIL-STD-810, except the test level and frequencv range shall be as specified in 3.3.4. Upon completion of the vibration cycling the Regulator shall be removed from the vibration table and adapter and subjected to the examinations in 4.5.6. Regulators that have passed the examinations specified in 4.5.6 shall then be subjected to the tests specified in 4,5.7 through 4.5.7.4.4 inclusive, to determine overall conformance to 3.3.4.

4.5.5 Vibration "B".- The test equipment utilized in this test shall conform to the requirements of 4.4.2.3.3. The regulator shall be vibrated in accordance with 3.3.5. Upon completion of the vibration cycling the Regulator shall be removed from the vibration table and adapter end subjected to the examinations in 4.5.6. Regulators that have passed the examinations specified in 4.5.6 shall then be subjected to the tests specified in 4.5.7 through 4.5.7.4.4 inclusive, to determine overall conformance to 3.3.5.

4.5.6 Examination.- Immediately following environmental test the Regulator shall be thoroughly examined for physical defects. A visual examination be made for broken leads, missing and damaged parts, materials, or finishes. A tactile examination shall also he conducted to determine looseness Of parts. Anyone of the possible defects defined above shall be cause to consider a Regulator defective.

4.5.7 Performance-Operational.- The following performance tests shall be conducted utilizing test eouipment specified in 4.4.2.3.4, and test loads as shown on Figure 1. Tests shall be performed with input voltages of 18.0 + 1.0 and 30.0 - 1.0 D.C. volts unless otherwise specified.

4.5.7.1 Polarization of input terminal.- When the voltages are applied as specified in 4.5.7 the input terminal shall be positive with respect to electrical circuit ground in compliance with 3.4.1.

4.5.7.2 Polarization of output terminals- When the voltages are applied as specified in 4.5.7 the output terminal shall be positive with respect to electrical circuit ground in compliance with 3.4.2.

4.5.7.3 Reverse polarity.- A reverse polarity of 30.0 -1.0 volts shall be applied for a period of five minutes between input terminal "D" and electrical circuit ground "F". Subsequent to this test the Regulator shall be subjected to all operational tests of 4.5.7.4 to determine compliance with 3.4.3.

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4.5.7.4 Intermittent Operation. - With 30.0 -1.0 volts applied to input terminal the Regulator shall be subjected to intermittent operation as specified in 3.4.4. Intermittent operation test to be conducted at room ambient temperature 75°F ± 15°F only. The Regualtor shall be subjected to the following tests subsequent to intermittent operation.

4.5.7.4.1 Input voltage surge.- Utilizing a pulse generator connected to the input terminal, superimpose voltage pulses as defined in 3.4.4.1 on the input potential of 30.0 -1.0 volts. A series of twelve pulses shall be applied with an interval of ten seconds between pulses. The output voltages, and recovery time after removal of pulses shall be as specified in 3.4.4.1.

4.5.7.4.2 Output current.- When the voltages are applied as specified in 4.5.7, the output current shall be as specified in 3.4.4.2 with the test shutter switch positioned in accordance with 3.4.4.4.

4.5.7.4.3 Output voltage.- When the voltages are applied as specified in 4.5.7, the output voltage and rise time shall be as specified in 3.4.4.3 with the test shutter switch positioned in accordance with 3.4.4.4.

4.5.7.4.4 Shutter switch operation.- The Regulator output shall be as specified in 3.4.4.4 with the test shutter switch positioned and input voltage conditions applied in the sequence noted below:

(1) Open test shutter switch
(2) Apply input voltage to regulator. Time shall be no output from Regulator.
(3) Close shutter switch. There shall be no output from Regulator
(4) Open shutter switch. Regulator output shall conform to 3.4.4.2 and 3.4.4.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet PS 11747171. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Regulator is a functional assembly of the passive Elbow Assembly for Periscopes: M35E1, M32E1, M36EI. The major function of this assembly is to provide regulated voltage.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Applicable packaging data sheet number (see 2.1)
(c) Selection of applicable levels of preservation, packaging and packing
(d) Applicable stock number
(e) Requirement for First Article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project Number 1240-A658

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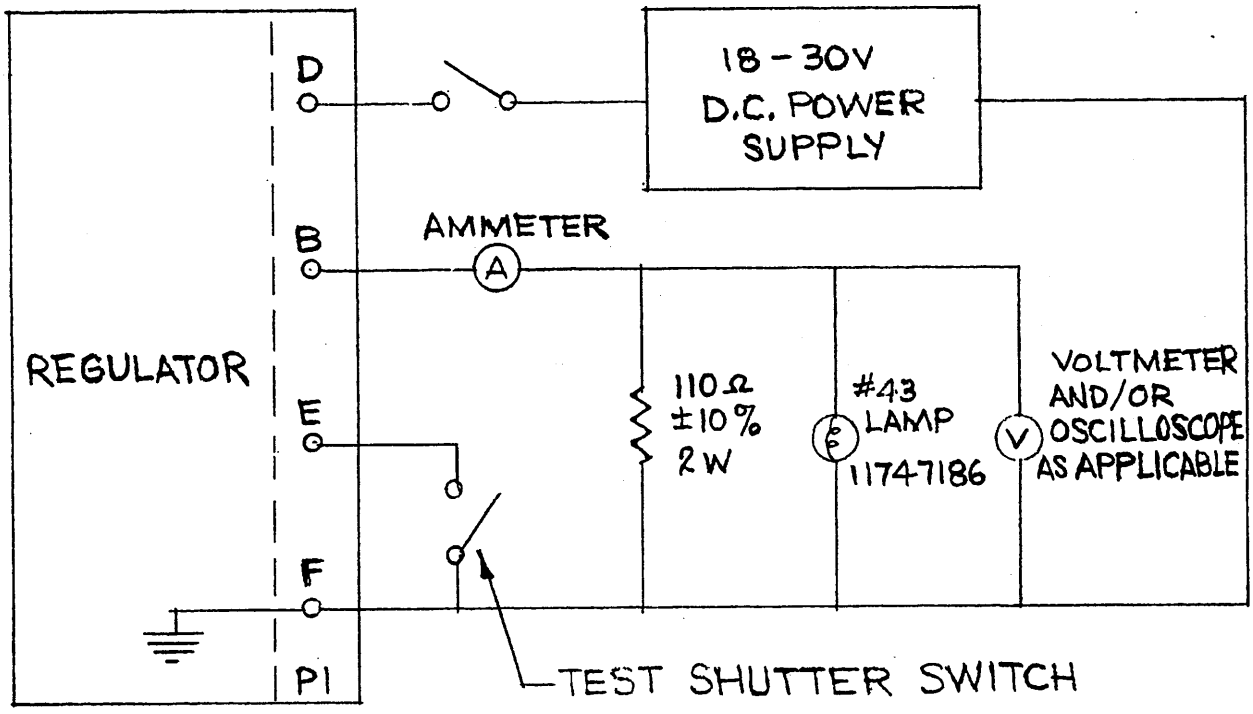


FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-R-48577(MU) Regulator Assembly: F11747171

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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MILITARY SPECIFICATION

REGULATOR ASSEMBLY: 11747171

This amendment forms a part of Military Specification MIL-R-48577 (MU) dated 27 June 1975.

PAGE 2

Paragraph 3.2.3, Delete and substitute:

"3.3.3 Shock.-The Regulator shall meet the requirements of 3.4 after be subjected to cumulative shocks consisting of 3 each at 100 G's and 40 G's applied in each direction along each of three mutually perpendicular axes. The shock pulse shall have a wave form of approximately a half sine wave. For 100 G's + 10%, the pulse shall have a time duration of 3 ± .2 milliseconds; for 40 G's + 10% the shock pulse shall have a time duration of 18 ± 3 milliseconds."

Paragraph 3.3.4, Delete and substitute:

"3.3.4 Vibration "A".-The Regulator shall meet the requirements of 3.4 after being vibrated sinusoidally along each of three mutually perpendicular axes as follows:

At a frequency of 5 Hertz (Hz), the acceleration shall be .5G ± 10%.

b. During a frequency sweep of 5 to 9 Hz, increase acceleration linearly from .5G to 1.5G's ± 10%.

During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G's ± 10%.

d. During a frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5 G's ± 10%.

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5 G's to .5 G ± 10%.

f. At 5 Hz, the acceleration shall be .5 G ± 10%.

FSC 1240

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The-frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis.

PAGE 3

Paragraph 3.3.5, Delete and substitute:

"3.3.5 Vibration "B".-This requirement shall be met with 24 ± 6 Vdc applied and an output current of 500 ma ± 25 ma. The vibration shall be sinusoidal as follows:

At a frequency of 5 Hertz (Hz), the acceleration shall be l.O G ± 10%.

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of 1.0 G ± 10%.

During a frequency sweep of 500 to 5 Hz, maintain constant acceleration of 1.0 G ± 10%.

d. At 5 Hz, the acceleration shall be 1.0 G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep for a total of 15 minutes ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes + 3 minutes). Following the vibration test, the regulator shall meet the requirements of 3.4."

Paragraph 3.4.4.1, Delete and substitute:

"3.4.4.1 Input voltage surge.-The regulator shall show no evidence of damage and the output voltage shall remain within 2.0 to 3.0 volts during the application of 100 millisecond positive square wave pulses having an amplitude of 80.0 ± 1.0 DC volts. The output voltage shall return to the level specified in 3.4.4.3 within 125 milliseconds after removal of pulses."

Paragraph 3.4.4.3, Delete and substitute:

"3.4.4.3 Output voltage.-The regulator shall deliver a dc voltage of 2.5 ± .3 volts with a rise time of 0.3 seconds or less upon opening test shutter switch."

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PAGE 10

Paragraph 4.5.7.4.1, Delete and substitute:

"4.5.7.4.1 Input voltage surge.-Utilizing a pulse generator connected to the input terminals, adjust the output current to 500 + 25 ma. Apply the pulses specified in 3.4.4.1. A seriees of five pulses shall be applied at a rate of five per minute. The output voltage and recovery time after removal of pulses shall be as specified in 3.4.4.1."

FIGURE 1

Delete and substitute new FIGURE 1.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1240-A728

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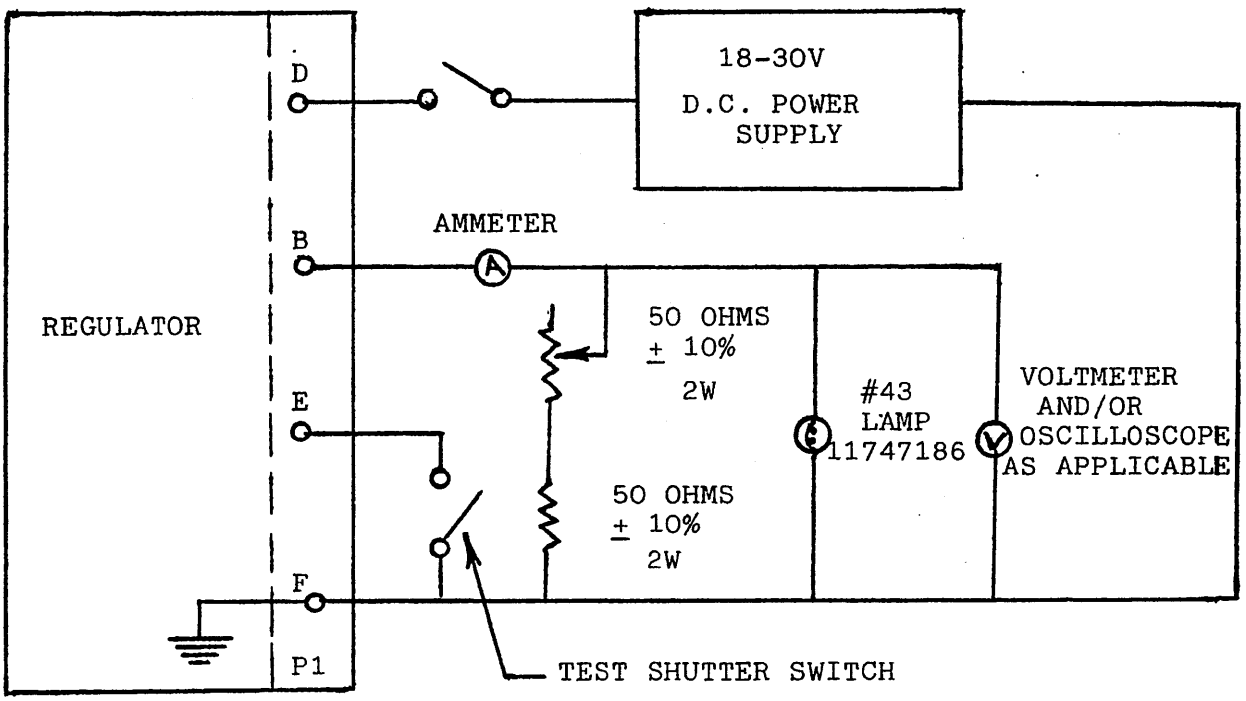


FIGURE 1

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